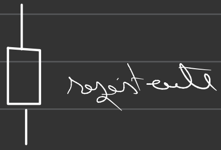
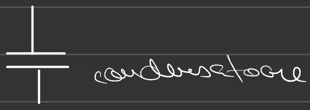
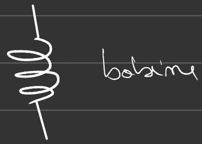
Curs 2


rezistențe


condensatoare



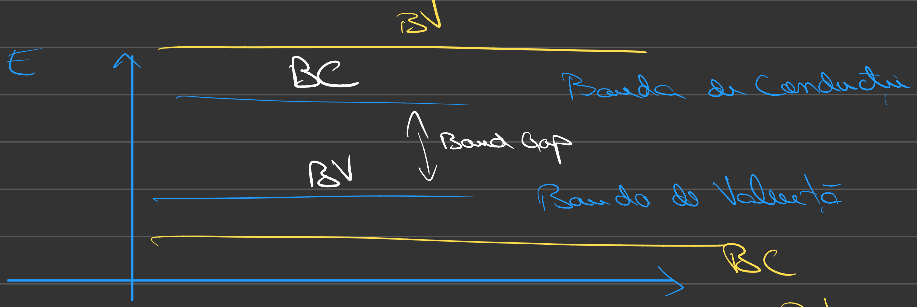
SEMICONDUCTORI

DIODA SEMICONDUCTOARE

izolatori (cauciuc, plastic, sticlă, lemn) -> band gap >> 15eV

conductori (cupru, metale)

semiconductori (siliciu, germaniu) -> band gap << 1eV


BV
E
BC
Banda de Conducție
Band Gap
BV
Banda de Valență
BC

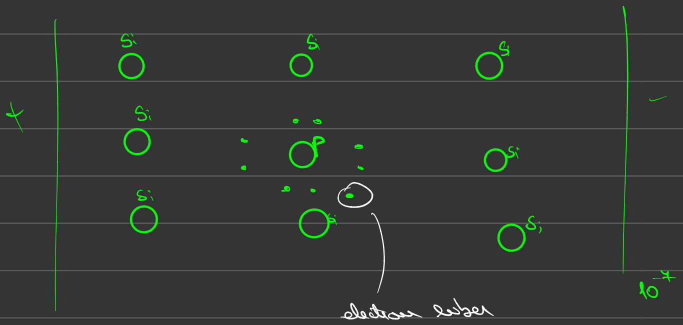
izolatori și semiconductori

Pentru metale se inversează

$$E = \frac{U}{R}$$

$$F \cong E_{g0}$$

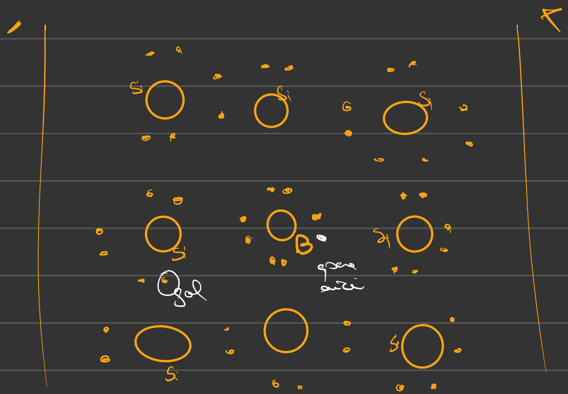
$h\nu$



Si
Si
Si
+
Si
P
Si
Si
Si
Si
-
10^7
electron liber

Curent de electroni (purtători negativi -)

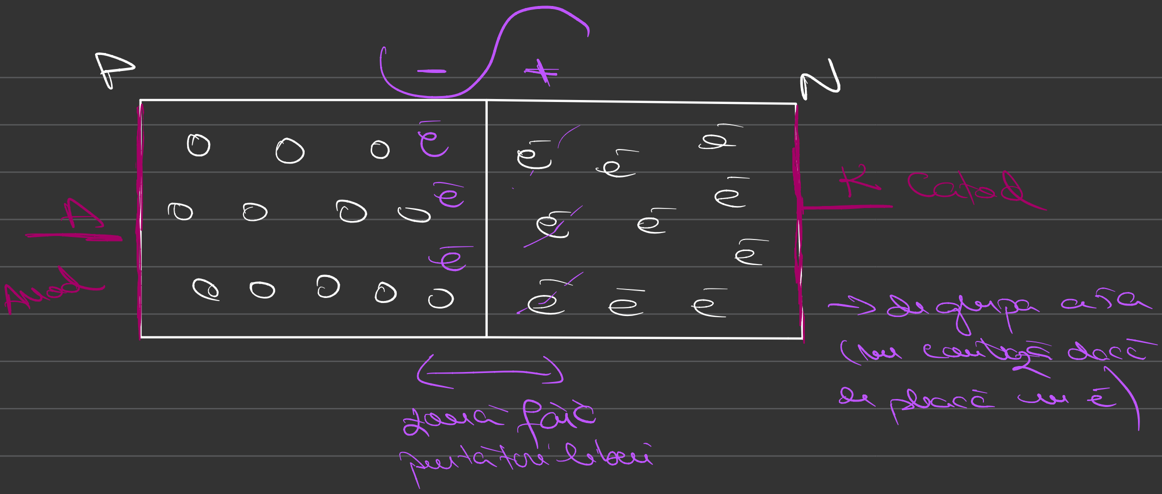
Semiconductor de tip N



+
Si
Si
Si
Si
B
Si
gol
spre aici
Si
Si

Curent de goluri (purtători pozitivi +)

Semiconductor de Tip P


P
N
−
+
Anod
A
K Catod
ē
ē
ē
ē
ē
ē
ē
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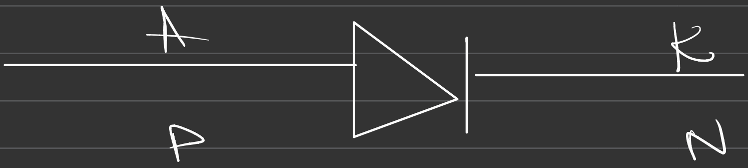
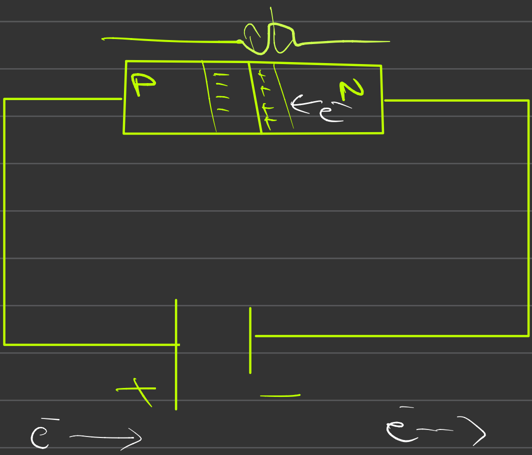
ē

→ de grupa asta (nu contează dacă se pleacă un ē)

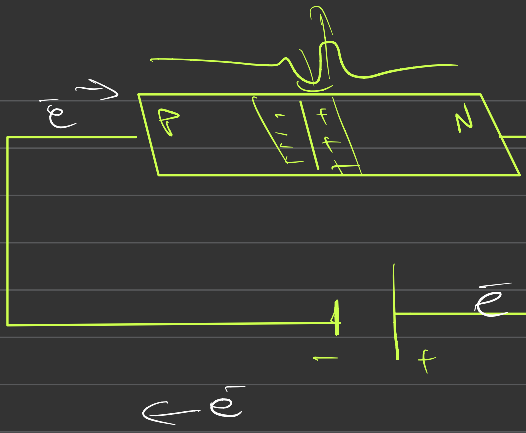
zonă fără purtători liberi





Polarizare directă

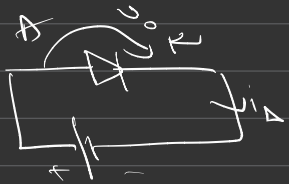
Curent de recombinare



Polarizare inversă

Curent de generare

P. directă

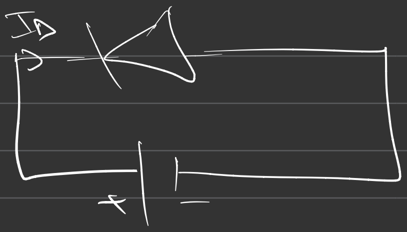


$$I_D = I_S\left(e^{\frac{v_D}{V_T}} - 1\right)$$

$$V_T = \frac{kT}{q_0} \cong 26\,mV$$

k – constanta lui K

T – temperatura absolută



P. inversă

$I_D = I_S$

Caracteristica Statică